LAPORAN PRAKTIKUM

SISTEM OPERASI

PERTEMUAN 9

(Syscall)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

Jurnal

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)

Open ▾ *chname.c ~/xinu/system Sav

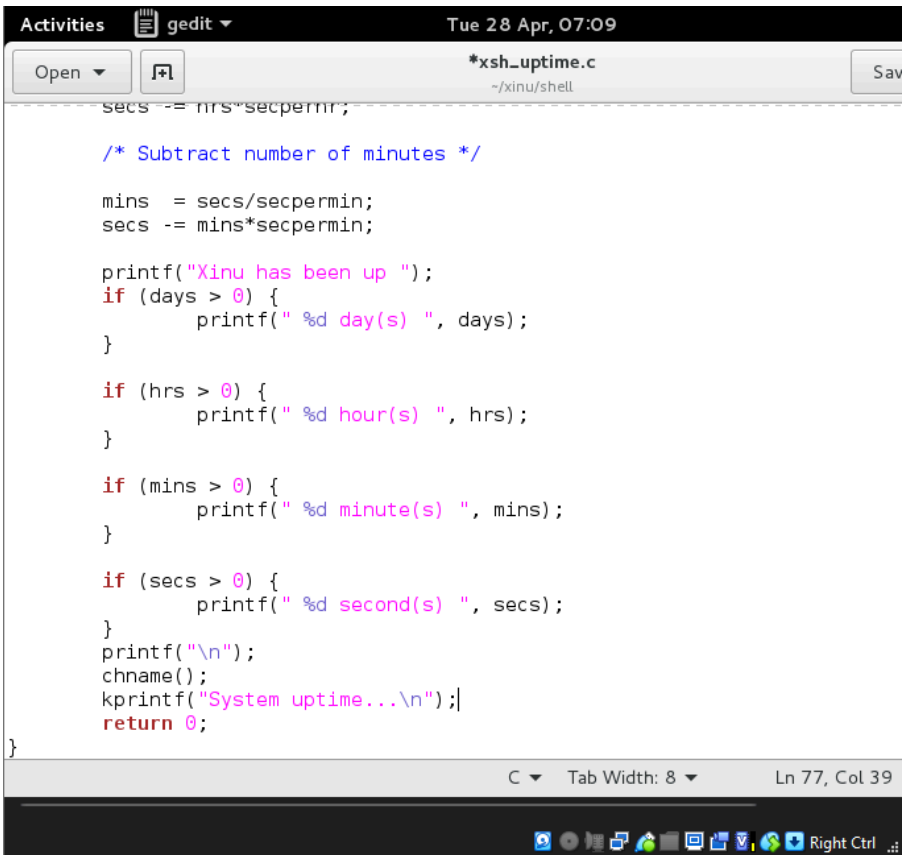
```c
/* chname.c - chname */

#include <xinu.h>
/* Nama: Izzaty zahara Br Barus
 * NIM : 2311104052
*/

syscall chname(void)
{
	intmask mask;
	mask = disable();
	kprintf("New Syscall is easy\n");
	restore(mask);
	return OK;
}
```

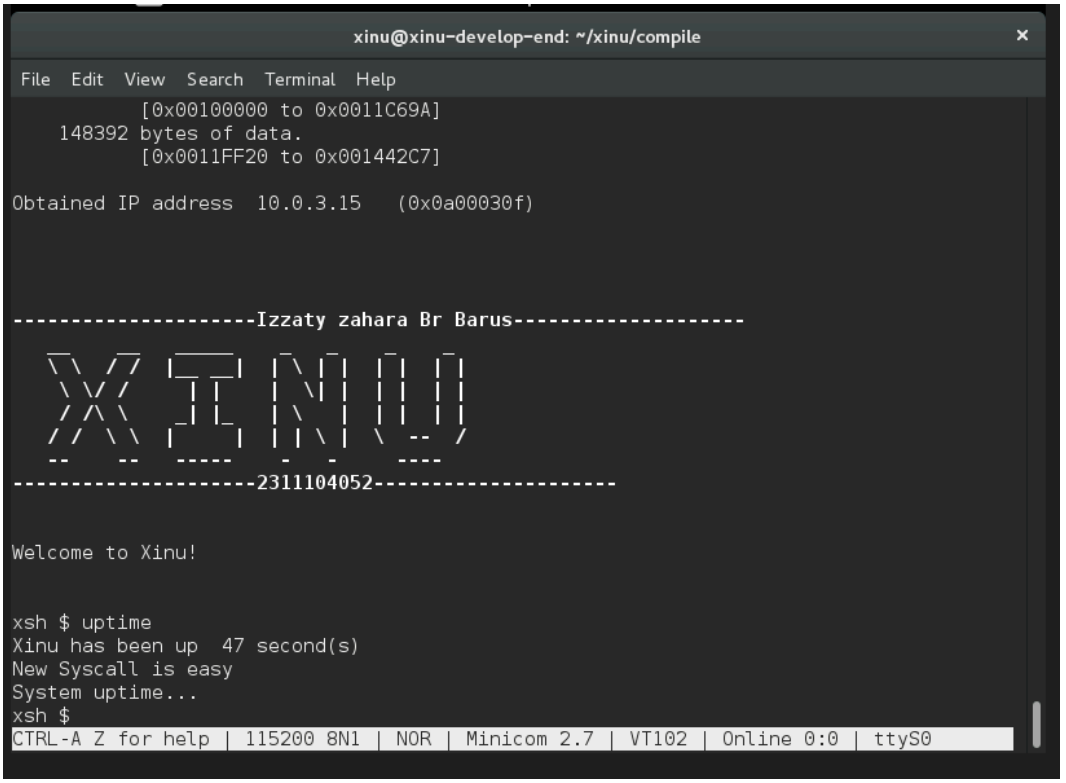
Activities gedit ▾ Tue 28 Apr, 07:10

Open ▾ prototypes.h ~/xinu/include Sav

```c
extern  void    arp_ntoh(struct arppacket *);
extern  void    arp_hton(struct arppacket *);

/* in file ascdate.c */
extern  status  ascdate(uint32, char *);

/* in file bufinit.c */
extern  status  bufinit(void);

/* in file chprio.c */
extern  syscall chprio(pid32, pri16);

/* in file chname.c */
extern  syscall chname(void);

/* in file clkupdate.S */
extern  uint32  clkcount(void);

/* in file clkhandler.c */
extern  interrupt clkhandler(void);

/* in file clkinit.c */
extern  void    clkinit(void);

/* in file clkdisp.S */
extern  void    clkdisp(void);

/* in file close.c */
```

C/C++/ObjC Header ▾ Tab Width: 8 ▾ Ln 28, Col 29





2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input
   - Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
   - Pastikan prioritas adalah bilangan yang positif Compile dan jalankan Xinu dengan syscall yang telah diperbaiki
   - make clean

- make

```c
/* chprio.c - chprio */

#include <xinu.h>

/*------------------------------------------------------------------------
 *  chprio  -  Change the scheduling priority of a process
 *------------------------------------------------------------------------
 */
syscall chprio(
          pid32          pid,            /* ID of process to change      */
          pri16          newprio         /* New priority                 */
        )
{
        intmask mask;                   /* Saved interrupt mask         */
        struct  procent *prptr;         /* Ptr to process's table entry */
        pri16   oldprio;                /* Priority to return           */

        mask = disable();
        if (isbadpid(pid)) {
                restore(mask);
                return SYSERR;
        }
        if (newprio <= 0){
                restore(mask);
                return SYSERR
        )
        prptr = &proctab[pid];
        oldprio = prptr->prprio;
```

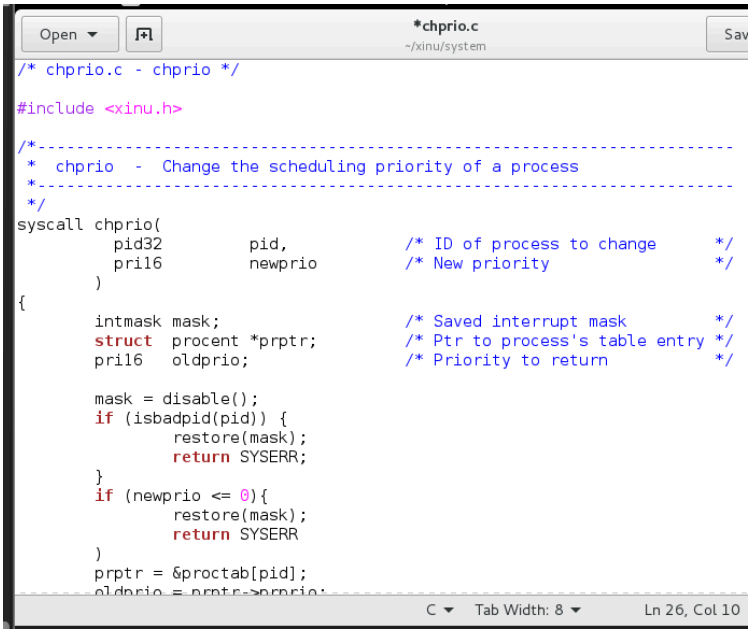
```
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFEF0     8192   1       3
  1 ipout             wait  500    0 0x005FBFFC 0x005FBF30     8192   0       0
  2 netin             wait  500    0 0x005F9FFC 0x005F9ED0     8192   0       0
  4 Main process      recv   20    3 0x005E7FFC 0x005E7F34    65536   0       0
  5 shell             recv   50    4 0x005D7FFC 0x005D79AC     8192   0       0
  6 ps                curr   20    5 0x005F7FFC 0x005F7DD8     8192   0       0
 xsh $ uptime
Xinu has been up  2 minute(s)  5 second(s)
New Syscall is easy
System uptime...
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFEF0     8192   1       3
  1 ipout             wait  500    0 0x005FBFFC 0x005FBF30     8192   0       0
  2 netin             wait  500    0 0x005F9FFC 0x005F9ED0     8192   0       0
  4 Main process      recv   20    3 0x005E7FFC 0x005E7F34    65536   0       0
  5 shell             recv   50    4 0x005D7FFC 0x005D79AC     8192   0       7
  8 ps                curr   20    5 0x005F7FFC 0x005F7FC4     8192   0       0
 xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0
```

3. Lakukan hal-hal berikut ini
   - Edit xsh_uptime.c Tambahkan kode berikut
   - Compile source code tersebut dengan perintah
   - make clean
   - make
   - Jalankan perintah ps

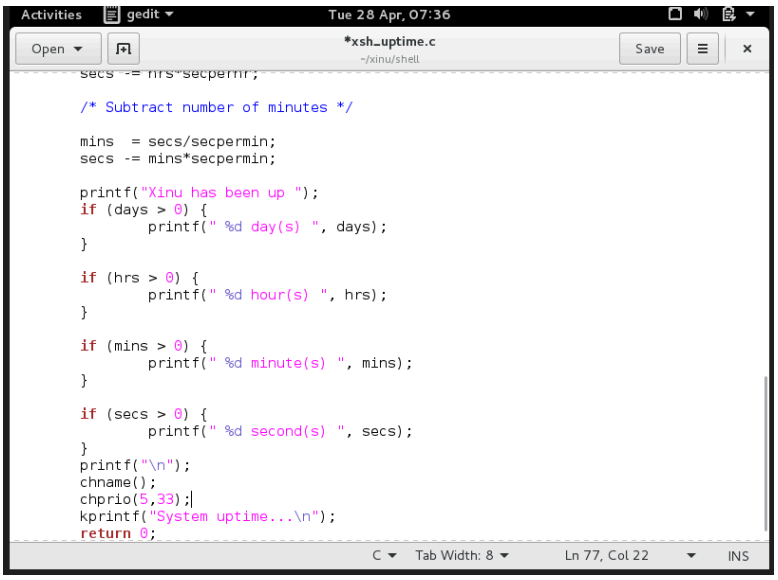
- xsh $ ps
- perhatikan prioritas proses dengan id = 5
- Jalankan uptime
- xsh $ uptime
- Perhatikan hasil perintah tersebut
- Jalankan ps
- xsh $ ps
- perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c
- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”
- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)




```
/* chprio.c - chprio */

#include <xinu.h>

/*------------------------------------------------------------------------
 *  chprio  -  Change the scheduling priority of a process
 *------------------------------------------------------------------------
 */
syscall chprio(
          pid32         pid,            /* ID of process to change      */
          pri16         newprio         /* New priority                 */
        )
{
        intmask mask;                   /* Saved interrupt mask         */
        struct  procent *prptr;         /* Ptr to process's table entry */
        pri16   oldprio;                /* Priority to return           */

        mask = disable();
        if (isbadpid(pid)) {
                restore(mask);
                return SYSERR;
        }
        if (newprio <= 0){
                restore(mask);
                return SYSERR
        )
        prptr = &proctab[pid];
        oldprio = prptr->prprio;
```




```
        secs -= hrs*secperhr;

        /* Subtract number of minutes */

        mins  = secs/secpermin;
        secs -= mins*secpermin;

        printf("Xinu has been up ");
        if (days > 0) {
                printf(" %d day(s) ", days);
        }

        if (hrs > 0) {
                printf(" %d hour(s) ", hrs);
        }

        if (mins > 0) {
                printf(" %d minute(s) ", mins);
        }

        if (secs > 0) {
                printf(" %d second(s) ", secs);
        }
        printf("\n");
        chname();
        chprio(5,33);
        kprintf("System uptime...\n");
        return 0;
```

```
Activities  Terminal   Tue 28 Apr, 07:38
xinu@xinu-develop-end: ~/xinu/compile
File  Edit  View  Search  Terminal  Help

Welcome to Xinu!


xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFEF0     8192   1       3
  1 ipout             wait  500    0 0x005FBFFC 0x005FBF30     8192   0       0
  2 netin             wait  500    0 0x005F9FFC 0x005F9ED0     8192   0       0
  4 Main process      recv   20    3 0x005E7FFC 0x005E7F34    65536   0       0
  5 shell             recv   50    4 0x005D7FFC 0x005D79AC     8192   0       0
  6 ps                curr   20    5 0x005F7FFC 0x005F7FC4     8192   0       0
 xsh $ uptime
Xinu has been up  New Syscall is easy
25 second(s)
System uptime...
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFEE4     8192   1       3
  1 ipout             wait  500    0 0x005FBFFC 0x005FBF30     8192   0       0
  2 netin             wait  500    0 0x005F9FFC 0x005F9ED0     8192   0       0
  4 Main process      recv   20    3 0x005E7FFC 0x005E7F34    65536   0       0
  5 shell             recv   33    4 0x005D7FFC 0x005D79AC     8192   0       7
  8 ps                curr   20    5 0x005F7FFC 0x005F7FC4     8192   0       0
 xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```